Shamima Hossain

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Research Interests

Understanding the capabilities of language models through their training data and learning dynamics; how models acquire knowledge, biases, and task-specific skills across stages of training; factuality and knowledge integration in vision–language models; interpretability of retrieval-augmented generation.

Education

Brac University *Dhaka, Bangladesh*
MSc in Computer Science (full time) Fall 2025 – present

- Research focus: how language models acquire knowledge, bias, and task-specific skills during different learning stages.

Brac University *Dhaka, Bangladesh*
BSc in Computer Science Jan 2017 – May 2021

- Undergraduate thesis: *Smart Companion Agent for Mental Well-being Using Deep Learning and NLP*.

Publications

[1] **Shamima Hossain**. *Beyond Generation: Multi-Hop Reasoning for Factual Accuracy in Vision–Language Models.* Accepted as a poster at the **ICML 2025** NewInML Workshop. [arXiv:2511.20531]

[2] **Shamima Hossain** et al. *Smart Companion Agent for Mental Well-being Using Deep Learning and NLP.* Undergraduate thesis. Proposed a non-invasive, transformer-based system (BioBERT) for real-time detection of mental distress from physiological signals, paired with a supportive conversational interface for continuous monitoring and early detection. [thesis]

Academic Service

MLSys 2026 — Ninth Annual Conference on Machine Learning and Systems
Program Committee Member, Artifact Evaluation Mar 2026 – Apr 2026

Research & Industry Experience

bKash Limited *Dhaka, Bangladesh*
Senior Research Engineer Jun 2023 – present

- Curated large-scale training data and built the training pipeline for AVA, bKash's first conversational agent — end-to-end ownership of data quality, filtering, and curation strategy for an LLM product serving a national-scale fintech user base.
- Designed a custom LLM-based dialogue system for financial transaction tasks, studying how task-specific grounding and dialogue structure affect reliability of user–agent interaction.
- Researched and implemented distributed training pipelines for efficient large-scale training of language and vision models.
- Developed and maintain a face anti-spoofing engine using deep vision methods, with emphasis on robustness and security under real-time constraints.
- Applied MLOps practices (AWS SageMaker, Lambda) for scalable training, inference, and deployment of production ML systems.

Research Projects

Identifying RAG Outputs in LLM-Generated Responses [code]

- Designed a strategy combining chain-of-thought reasoning and token manipulation that lets users visually identify retrieved external knowledge within LLM outputs, improving transparency and interpretability of RAG systems.

- Demonstrated that LLMs can self-identify gaps in their parametric knowledge and tag externally retrieved information — evidence of capabilities relevant to studying what models learn from training data versus context.

Factual Image Captioning with Vision–Language Models and Knowledge Graphs [code]

- Ran controlled experiments on hallucination in VLMs by injecting hierarchical structured knowledge (country → city → landmark) to improve factual accuracy in image captioning.
- Systematically compared hierarchical vs. unstructured knowledge augmentation using Qwen2-VL-2B-Instruct and GraphX, showing hierarchical structure notably improves VLMs' ability to fact-check visual content.

Extracting Reusable Filters from Diffusion Models for Controlled Image Manipulation [code]

- Engineered a method to compute semantic difference matrices from text embeddings in diffusion models, enabling visual exploration of prompt-driven transformations in latent and pixel space.
- Showed prompt-difference matrices map to transferable semantic shifts across images, contributing to interpretability of latent-space dynamics.

Prompt Autocomplete: End-to-End ML Workflow Pipeline [code]

- Built and deployed a production-ready pipeline for training and scaling transformer models for prompt autocompletion in text-to-image applications (AWS SageMaker, Lambda, API Gateway, Hugging Face PyTorch).

Open-Source Contributions

Hugging Face *Remote*
Contributor 2022 – present

- Contributed to zero-shot image classification documentation in `transformers`.
- Fine-tuned Stable Diffusion models with DreamBooth for the Keras DreamBooth community event.

Volunteering

United Nations OCHA *Remote*
Junior Data Analyst & Web Designer May 2020 – Sep 2020

- Supported data preparation and visualization for GIMAC Ethiopia and Iraq policy analysis during the COVID-19 outbreak; designed components of the GIMAC website and a data-analysis portal mock-up.

Technical Skills & Certifications

Languages	Python, C
Frameworks	PyTorch, AutoGen, DSPy, Django, FastAPI
Tools	Git, AWS (SageMaker, Lambda), VS Code, IntelliJ
Certifications	AWS Machine Learning Engineer Nanodegree, Generative AI, Deep Learning (Udacity)